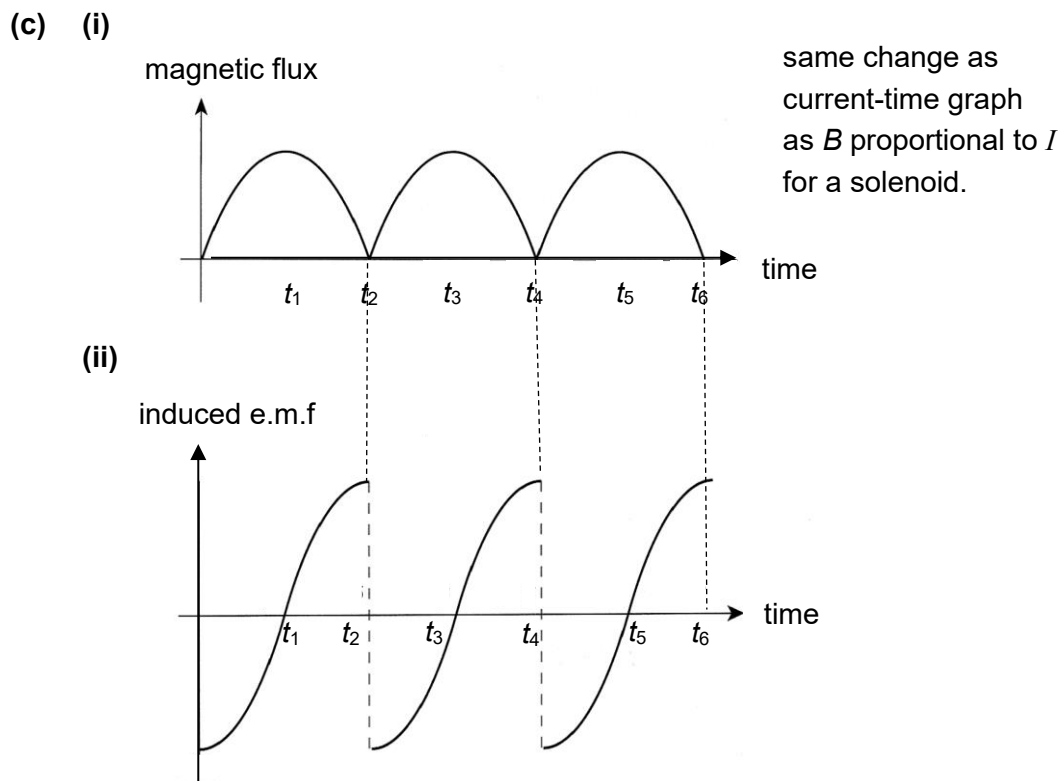


5	(a)	(i) This increases the magnetic flux linkage with secondary coil (ii) This reduces heat/energy losses caused by eddy currents in the core	B1 B1

- (b) The changing current in the primary coil gives rise to changing magnetic field and hence changing magnetic flux in the iron core. **B1**
The iron core links the magnetic flux in the primary to the secondary coil. **B1**
From Faraday's law, the changing flux in the secondary coil induces e.m.f. in secondary coil. **B1**

MC:	Generally poorly attempted. Some poor answers include: changing current causes changing magnetic flux and general statement of Faraday's law without applying it in context to the question does not gain credit.
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MC:	According to Faraday's law, e.m.f. induced in secondary coil = rate of change of magnetic flux linkage in secondary coil or $\varepsilon \propto \frac{d\Phi}{dt}$ Assuming number of turns is same for both primary and secondary coil, then flux linkage is the same in both coils. Graph of e.m.f. is the negative gradient of the magnetic flux-time graph.
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- (iii) 1 If the number of turns in the primary coil increases, the e.m.f. induced in the secondary coil decreases, assuming the number of turns in the secondary coil remains the same. **B1**
 This is because the e.m.f. induced in the secondary coil is inversely proportional to the number of turns in the primary coil when the input p.d. also remains constant. **B1**
- 2 If the number of turns in the secondary coil increases, the e.m.f. induced across the secondary coil increases, assuming the number of turns in the primary coil remains constant. **B1**
 This is because the e.m.f. induced in the secondary coil is directly proportional to the number of turns in the secondary coil when the input p.d. also remains constant. **B1**

MC:	Generally poorly done. Answers to (c)(iii) can easily be understood from the ideal transformer equation
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